

Chapter 1

Introduction

In our ever evolving technological world, new ways of solving problems are being developed at a rapidly increasing rate. There is much to be said about developing brand-new methods in solving existing problems, however, in this report I propose the combination of two, fairly well-thought-out existing technologies, to try and solve a useful problem in the world of [Radio Detection and Ranging \(RADAR\)](#).

This report outlines the thought process, design, and implementation of a single pulse RADAR system, combined with an [Orthogonal Frequency Division Multiplexing \(OFDM\)](#) based signal, to achieve the goal of target classification from a scene. RADAR systems in today's world can perform certain tasks extremely well (in the right conditions), however, there still seems to be a lack of RADAR systems that can identify/classify what objects might be sound in a scene of interest, without the utilization of complex image processing techniques.

In this paper, I will outline the process that was gone through to build a working prototype system, that uses the combination of a single pulse RADAR, and an OFDM based signal, to try and classify objects in a scene.

This report will consist of the following loose structure. Firstly, a literature review of the related works to do with the fundamentals of the project, namely, single pulse RADARs, OFDM, and the mixture of the two. Next, a theory section that concerns the workings behind the project will be presented. Thirdly, the design of the surrounding system to transmit and receive data, process the data, and analyze the results is presented. Following this, an experimental design is laid out and explained in detail. This is followed by the performing of real world testing which is then analyzed in the results section. Lastly, there will be some discussion of results and further steps that could be implemented.

1.1 Background

RADAR technology has unlocked the ability for humans to be able to see the world around us with a new perspective. This ability to be able to see objects and more specifically, to see information such as position and velocity about these objects has helped with the development of devices and services that are of paramount importance to society. All the current use cases of RADAR do however lack the ability to determine exactly what object is being seen by the overall RADAR system. That is where the use of the OFDM waveform in a RADAR system becomes useful.

1.2 Problem Statement

Current RADAR technology works very well when determining the distance and direction of an object, however, it does not cope well with specific object classification tasks. One of the reasons for this is that current RADAR utilize either a pulse, continuous wave, or chirp waveform, which does not allow for object detection through statistical analysis due to the simple nature of the waveform. This is a drawback of using RADAR systems, since if object classification is desired, then a combination of radar and some other technology such as cameras would have to be used which increases the cost and complexity of the system as a whole.

1.3 Objectives

The goal of this project is to build a RADAR system that can simultaneously act as a normal RADAR which can determine the distance and direction of an object, as well as classify the specific object that it is looking at when compared to other example objects. The aim is to implement an OFDM based waveform into a RADAR system that would give the flexibility to analyze the returned signal statistically and determine what exactly the object is. The goal is to make a non-realtime system, that can distinguish between three different objects.

1.4 System Requirements

The RADAR project has the following requirements:

1.4.1 Hardware Requirements

The main hardware for the project should be widely available computer scientific equipment. The availability of computer equipment is generally larger than most of the other required equipment such as the [Software Defined Radio \(SDR\)](#) equipment. The SDR equipment is much less widely available due to the high price of the equipment itself. This is fine however, since the [University of Cape Town \(UCT\)](#) has in their control multiple [Universal Software Radio Peripheral \(USRP\)](#) N210 devices designed and manufactured by Ettus research which will be perfect for my project.

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1. Be able to operate as a normal RADAR. This entails determining range, azimuth, and elevation.
2. Be able to classify between three separate reflectors/scatterers.

1.5 Scope and Limitations

The scope of this project has been set to not include design of RADAR hardware specifically. This is because design of RADAR hardware is a complex task that would probably encompass an entire masters project in and of itself. Therefore, the RADAR hardware portion of the project will be handled by a [SDR](#). This extrapolates the complex RADAR hardware design aspect away from the project and allows me to focus on the data collection and processing aspect of the project.

Bibliography

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Glossary

OFDM Orthogonal Frequency Division Multiplexing

RADAR Radio Detection and Ranging

SDR Software Defined Radio

UCT University of Cape Town

USRP Universal Software Radio Peripheral